

Bertan Karacora

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EDUCATION

M.Sc. in Computer Science

Oct. 2022 – Present

University of Bonn

Bonn, Germany

- Current grade average: 1.1/1.0 (based on degree requirements)
- Focus areas: Machine Learning, Computer Vision
- Thesis pending, expected completion: 2026

B.Sc. in Computer Science

Oct. 2018 – Sep. 2022

University of Bonn

Bonn, Germany

- Final grade: 1.7/1.0
- Thesis: *Inverse Rendering of Wave Optical BRDFs*, grade: 1.1/1.0, supervisor: Prof. Matthias Hullin

General Qualification for University Entrance

Jul. 2010 – Jun. 2018

Carl-Fuhlrott-Gymnasium

Wuppertal, Germany

- Final grade: 1.4/1.0
- Awarded the MINT-EC Certificate with distinction

EXPERIENCE

Student Teaching Assistant

Oct. 2025 – Present

Computational Analytics Group, University of Bonn, Prof. Estela Suarez

Bonn, Germany

- Served as a tutor for *Introduction to High Performance Computing*

Student Research Assistant

Apr. 2024 – Sep. 2025

Autonomous Intelligent Systems Group, University of Bonn, Prof. Sven Behnke

Bonn, Germany

- Contributed to the NimbRo team's participation in RoboCup@Home competitions for autonomous service robots:
 - * RoboCup 2024 (Eindhoven, Netherlands) – 1st place
 - * RoboCup German Open 2025 (Nuremberg) – 1st place
 - * RoboCup 2025 (Salvador, Brazil) – 2nd place
- Integrated a fisheye-lens camera for large-FOV perception into the NimbRo robot, including projection model selection, intrinsic/extrinsic calibration, image rectification, and LiDAR point cloud colorization
- Developed methods for large-FOV depth estimation by combining dense monocular depth estimation with sparse 3D LiDAR measurements (e.g., finetuning using sparse supervision)
- Handled problems of the existing NimbRo robot's person detection and face recognition components, and incorporated person re-identification using deep embeddings of the full-body appearance
- Enabled continuous person detection during robot movements using a deep learning approach for omnidirectional person detection in 2D LiDAR scans
- Developed a 3D multi-object tracking framework compatible with the Nimbro robot's sensor setup, based on Kalman filtering, a deep association metric, and uncertainty-aware track management
- Implemented the RoboCup@Home task *Help Me Carry*, demonstrating human-robot interaction, person tracking and following in crowded environments, grasping and carrying of bags, navigation and obstacle avoidance

Student Teaching Assistant

Oct. 2022 – Mar. 2023

Technical Computer Science Group, University of Bonn, Prof. Joachim Anlauf

Bonn, Germany

- Served as a tutor for *Technical Informatics*

Working Student - Full-Stack Developer

Sep. 2020 – Sep. 2022

Pixelrein GmbH & Co. KG

Bonn, Germany

- Extended functionalities of a web application for an SEO services marketplace

SELECTED PROJECTS AND WRITING SAMPLES

Underlined text indicates links to public repositories in my GitHub profile and PDFs on my personal website.

Learning Context-Aware 3D Scene Representations with Graph Autoencoders 2025

Guided research project at the Autonomous Intelligent Systems Group | [Code](#)

- Developed a GNN-based autoencoder to learn context-enriched object embeddings from 3D scene graphs
- Adopted an optimal transport loss to align real and reconstructed feature distributions of neighboring objects, enabling task-agnostic self-supervised pretraining
- Incorporated the pretrained encoder into a custom 3D visual grounding model and demonstrated its effectiveness on the NR3D dataset, indicating potential for state-of-the-art performance with full-scale training and optimization

Surface-Aligned Gaussian Splatting for Efficient 3D Mesh Reconstruction and Rendering 2024 – 2025

Seminar in Recent Advances in Geometry Processing | [Report](#), [Slides](#)

Protein Classification Challenge 2025

Course project in Geometric Deep Learning | [Code](#)

- Participated in a Kaggle competition on protein classification using GNNs

The Impact of Fiber Orientation Features for White Matter Tract Segmentation 2023 – 2024

Lab project in Visualization and Medical Image Analysis | [Report](#), [Slides](#), [Code](#)

- Reproduced the results of TractSeg, a U-Net-based segmentation model for tract segmentation in dMRI images
- Adapted the data processing and training of TractSeg to replace voxel-wise peak vectors with higher-order tensor and low-rank representations of the fiber orientation distribution
- Assessed the impact of different input features on TractSeg's performance quantitatively and qualitatively

Neural Parametric Models for 3D Deformable Shapes 2023

Seminar in Visual Computing | [Report](#), [Slides](#)

Inverse Rendering of Wave Optical BRDFs 2021 – 2022

Bachelor's thesis | [Thesis](#), [Slides](#), [Code](#)

- Investigated differentiable rendering for reconstructing specular surface microgeometry from BRDF slices while accounting for wave-optical effects

Computer Animation Festival 2021

Course project in Introduction to Computer Graphics and Visualization | [Code](#)

- Developed a video game prototype and showcased implemented concepts from the lecture

Lightweight Multi-Branch Network for Animal Re-Identification 2021

Lab project in Intelligent Vision Systems | [Report](#), [Slides](#), [Code](#)

- Created a wild animal dataset from camera-trap videos via semi-supervised annotation
- Trained and evaluated LightMBN for animal re-identification, incorporating varied frame-sampling and cropping strategies to handle video-based data

PUBLICATIONS

RoboCup@Home 2024 OPL Winner NimbRo: Anthropomorphic Service Robots using Foundation Models for Perception and Planning. R. Memmesheimer, J. Nogga, B. Pätzold, E. Kruzhkov, S. Bultmann, M. Schreiber, J. Bode, **B. Karacora**, J. Park, A. Savinykh, S. Behnke. RoboCup 2024: Robot World Cup XXVII, LNCS 15570, Springer, 2025.

SKILLS

Languages: German (native), English (fluent), French (basic)

Programming: Python (proficient), C/C++ (intermediate), Matlab (familiar)

Frameworks: Pytorch, Pytorch Geometric, TorchVision, CUDA, OpenGL

Developer Tools: Git, VS Code, Jupyter Notebook, Docker, Bash scripting, Linux system

Miscellaneous: TensorBoard, ONNX, TensorRT, ROS 2, L^AT_EX